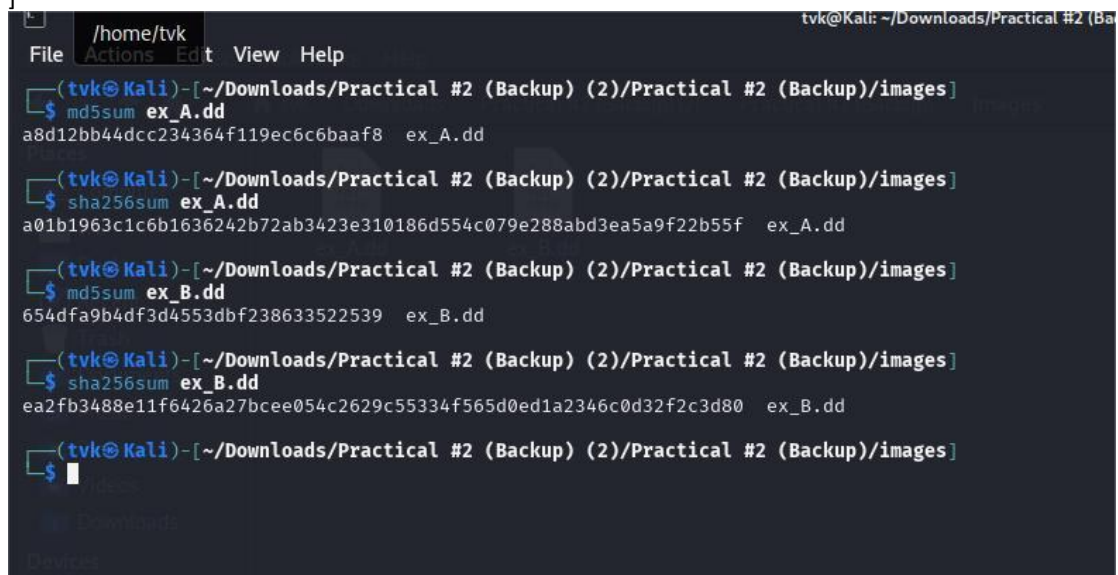


Practical Sheet 2
Data Manipulation

1. Give the MD5 and SHA256 hashes of both images.

Ans:[

```
{
  "filename": "ex_A.dd",
  "md5": "a8d12bb44dcc234364f119ec6c6baaf8",
  "sha256": "a01b1963c1c6b1636242b72ab3423e310186d554c079e288abd3ea5a9f22b55f"
},
{
  "filename": "ex_B.dd",
  "md5": "654dfa9b4df3d4553dbf238633522539",
  "sha256": "ea2fb3488e11f6426a27bcee054c2629c55334f565d0ed1a2346c0d32f2c3d80"
}
]
```



The screenshot shows a Kali Linux terminal window with the following commands and output:

```
tvk@Kali: ~/Downloads/Practical #2 (Backup) (2)/Practical #2 (Backup)/images
$ md5sum ex_A.dd
a8d12bb44dcc234364f119ec6c6baaf8  ex_A.dd

$ sha256sum ex_A.dd
a01b1963c1c6b1636242b72ab3423e310186d554c079e288abd3ea5a9f22b55f  ex_A.dd

$ md5sum ex_B.dd
654dfa9b4df3d4553dbf238633522539  ex_B.dd

$ sha256sum ex_B.dd
ea2fb3488e11f6426a27bcee054c2629c55334f565d0ed1a2346c0d32f2c3d80  ex_B.dd
```

In Kali Terminal I ran the following commands:

```
md5sum ex_C.dd
sha256sum ex_C.dd
md5sum ex_D.dd
sha256sum ex_D.dd
```

2. Give details on all allocated partitions in both images. Disregard unallocated/deleted partitions.

Ans:{

```
"ex_A.dd": [
  {
    "start-sector": "63",
    "end-sector": "3140927",
    "size": "1608017280",
    "partition-type": "0x06",
    "partition-type-name": "FAT16"
  }
]
```

```

],
"ex_B.dd": [
{
  "start-sector": "2048",
  "end-sector": "12580863",
  "size": "6440562688",
  "partition-type": "0x83",
  "partition-type-name": "Linux"
}
]
}

```

I've used mmls to display the contents of the partition table based on the results I got I created this JSON.

```

(tvck@Kali)-[~/Downloads/Practical #2 (Backup) (2)/Practical #2 (Backup)/images]
$ mmls ex_A.dd

DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length    Description
000:  Meta      0000000000 0000000000 0000000001 Primary Table (#0)
001:  _____ 0000000000 0000000062 0000000063 Unallocated
002:  000:000 0000000063 0003140927 0003140865 DOS FAT16 (0x06)
003:  _____ 0003140928 0003145727 00000004800 Unallocated

(tvck@Kali)-[~/Downloads/Practical #2 (Backup) (2)/Practical #2 (Backup)/images]
$ mmls ex_B.dd

DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

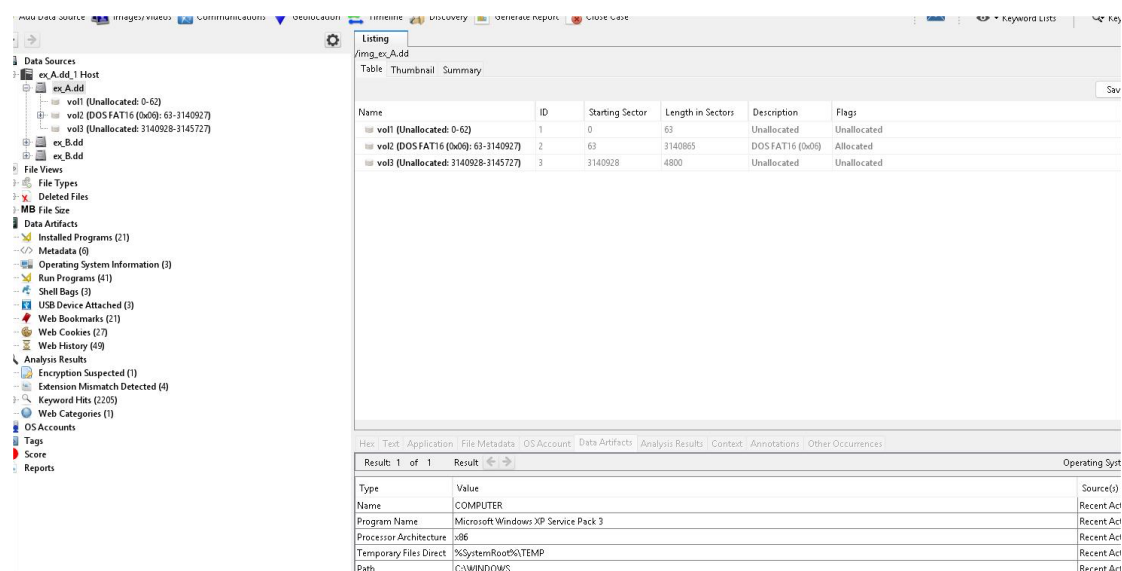
Slot      Start      End      Length    Description
000:  Meta      0000000000 0000000000 0000000001 Primary Table (#0)
001:  _____ 0000000000 0000002047 0000002048 Unallocated
002:  000:000 0000002048 0012580863 0012578816 Linux (0x83)
003:  _____ 0012580864 0012582911 0000002048 Unallocated

(tvck@Kali)-[~/Downloads/Practical #2 (Backup) (2)/Practical #2 (Backup)/images]
$

```

3. Pick the operating system that was installed on the computer system when creating the corresponding images. You get 0.5 points for each correct choice.

Ans: a) For ex A.dd
Windows XP service Pack 3



b) For ex B.dd
Ubuntu 14.04*LTS

```
cat /mnt/ex_mount/etc/os-release
cat /mnt/ex_mount/etc/issue
```

I used these commands on a mounted drive to get the information from the os-release file

```
sudo mount -o io,loop,offset=$((2048*512)) ex_B.dd /mnt/ex_mount
ls /mnt/ex_mount
bin boot cdrom dev etc home initrd.img lib lost+found media mnt opt proc root run sbin srv sys tmp usr var vmlinuz
(tvk@Kali)-[~/Downloads/Practical #2 (Backup) (2)/Practical #2 (Backup)/images]
→ cat /mnt/ex_mount/etc/os-release
cat /mnt/ex_mount/etc/issue
NAME="Ubuntu"
VERSION="14.04.1 LTS, Trusty Tahr"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 14.04.1 LTS"
VERSION_ID="14.04"
HOME_URL="http://www.ubuntu.com/"
SUPPORT_URL="http://help.ubuntu.com/"
BUG_REPORT_URL="http://bugs.launchpad.net/ubuntu/"
Ubuntu 14.04.1 LTS \n \l
```

4. a) Identify all user accounts (those that are intended to be used by individuals, not service accounts). Give the password hashes of the accounts (just the hash). If there are multiple hash values, take the first. Crack the passwords using a tool we have discussed in the exercise session. To be able to do that, you might need to use another tool, for example one that is available in Kali's tool list. Also indicate if the user has root/administrator privileges ("root" or "non-root")

Ans: {

```
"ex_A.dd": [
{
  "username": "Administrator",
  "hash": "31d6cfe0d16ae931b73c59d7e0c089c0",
  "password": "",
  "privilege-level": "root"
},
{
  "username": "Gast",
  "hash": "31d6cfe0d16ae931b73c59d7e0c089c0",
  "password": "",
  "privilege-level": "non-root"
},
]
```

```
{
  "username": "John",
  "hash": "31d6cfe0d16ae931b73c59d7e0c089c0",
  "password": "",
  "privilege-level": "non-root"
}
],
"ex_B.dd": [
  {
    "username": "frank",
    "hash":
"$6$wEcBhZdp$DOa/QsIJ2vVMfInSE.4xWfo5bRScglBUJFJSWXgs7eiJhI0Fu9om7EuZwKRNip1oDZeliXC
1L3vrEp4MzYPC0",
    "password": "frank",
    "privilege-level": "root"
  }
]
}
```

For ex A.dd I extracted SAM and SYSTEM files and moved those to Kali after preparing the folder I used NT hash mode to crack the passwords using John.

`john --format=NT exA_hashes.txt --wordlist=/usr/share/wordlists/rockyou.txt`

```
(tvk@Kali)-[~/exA_hash]
$ john --format=NT exA_hashes.txt --wordlist=/usr/share/wordlists/rockyou.txt

Using default input encoding: UTF-8
Loaded 3 password hashes with no different salts (NT [MD4 128/128 SSE2 4x3])
Warning: no OpenMP support for this hash type, consider --fork=8
Press 'q' or Ctrl-C to abort, almost any other key for status
(Administrator)
1g 0:00:00:00 DONE (2025-06-08 01:13) 1.190g/s 17075Kp/s 17075Kc/s 34157KC/s markinho..*7;Vamos!
Warning: passwords printed above might not be all those cracked
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

(tvk@Kali)-[~/exA_hash]
$ john --show --format=NT exA_hashes.txt

Administrator::500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
*disabled* Gast::501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
John::1003:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::

3 password hashes cracked, 2 left
```

For ex B.dd

I ran John the Ripper to crack a password hash in combined.txt using the rockyou.txt wordlist.

I used grep frank shadow to get the hash as the shadow file contains all the hash passwords.

```

(tvtk@Kali)-[~/exB_hash]
$ john combined.txt --wordlist=/usr/share/wordlists/rockyou.txt
john --show combined.txt

Using default input encoding: UTF-8
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 128/128 SSE2 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
frank (frank)
1g 0:00:00:00 DONE (2025-06-08 01:26) 1.298g/s 2659p/s 2659c/s 2659C/s clover..lovers1
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
frank:frank:1000:1000:Frank Doe,,,:/home/frank:/bin/bash

1 password hash cracked, 0 left

(tvtk@Kali)-[~/exB_hash]
$

(tvtk@Kali)-[~/exB_hash]
$ cat /mnt/ex_mount/etc/group | grep sudo
sudo:x:27:frank

```

4b) From the analysis we got that password matches user's real name which was Frank Doe.

5. Which programs/packages were manually installed by users on the two computers? Exclude programs that were pre-installed or are an integral part of the operating system. For Windows, give the filename of the executable. For Linux, give the package name. You get 0.75 points for each image.

```

{
  "ex_C.dd": ["program1.exe", "program2.exe", "program3.exe", ...],
  "ex_D.dd": ["python3", "latex", "tamarin-prover", ...]
}

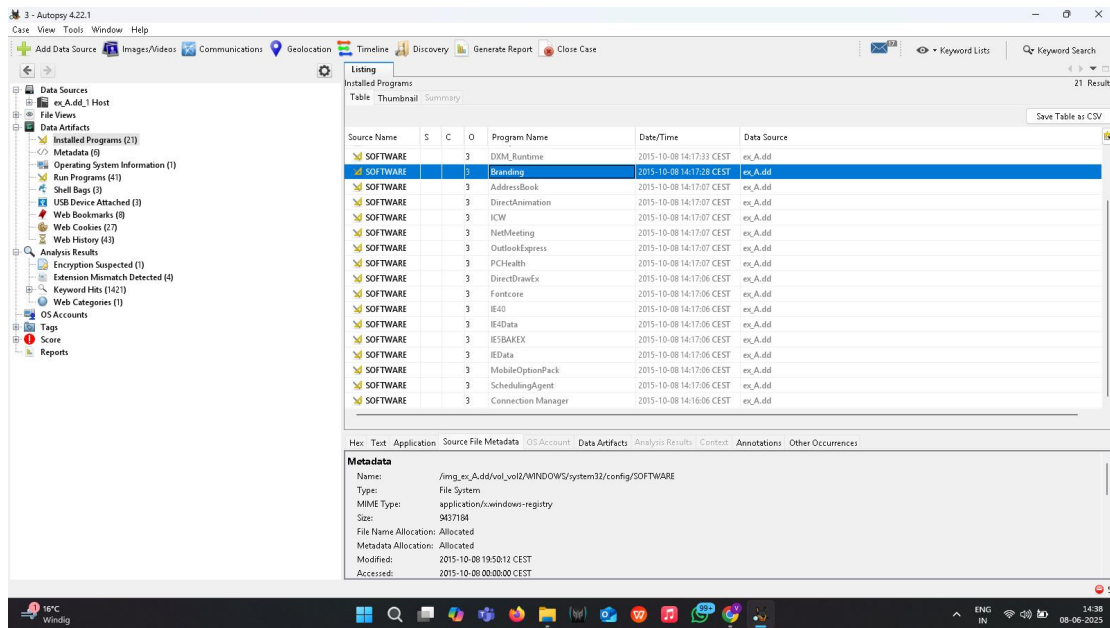
```

```

{
  "ex_A.dd": ["firefox.exe", "maintenanceservice.exe", "mplayer2.exe"],
  "ex_B.dd": ["apg", "baobab"]
}

```

I went through the File System in autospy and went to Installed apps section:



7. For each of the images (= hard drive duplicates), name five relevant log files. Give the full path. A log file is considered relevant if it contributes information to the questions asked by the prosecutor's office. Do not give more than five files. You get 1 point for five valid files, 0.5 if at least 3 files are relevant.

```

{
  "ex_C.dd": [ "C:/Windows/.../log1.log", "D:/Backup/something.log", ... ],
  "ex_D.dd": [ "/path/to/file1.log", "path/to/another_file.log", ... ]
}

```

```

Ans: {
  "ex_A.dd": [
    "C:\\WINDOWS\\Debug\\NetSetup.log",
    "C:\\WINDOWS\\Prefetch\\FIREFOX.EXE-1D57670A.pf",
    "C:\\WINDOWS\\WindowsUpdate.log",
    "C:\\Dokumente und Einstellungen\\John\\NTUSER.DAT.LOG",
    "C:\\WINDOWS\\Debug\\blastcln.log"
  ],
  "ex_B.dd": [
    "/var/log/auth.log",
    "/var/log/dpkg.log",
    "/var/log/syslog",
    "/var/log/apt/history.log",
    "/home/frank/.bash_history"
  ]
}

```

In autopsy I went through the file system to check the prefetch and WINDOWS\system32\LogFiles, WINDOWS\Debug to get the log files.

